

EION MULRENIN

Emory University
Department of Mathematics
Math & Science Center, N404

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<http://www.emulrenin.github.io/>

ACADEMIC INTERESTS

Ramsey theory, extremal combinatorics and graph theory, pseudorandom graphs and finite geometry.

EDUCATION

Emory University

- Ph.D., Mathematics (expected graduation: 2028)
 - Advisor: Cosmin Pohoata

University of South Florida

- B.A., Pure Mathematics, *summa cum laude* (2023).
 - Advisor: Brendan Nagle
- B.A., Philosophy, *summa cum laude* (2023).

PUBLICATIONS AND PREPRINTS

5. D. Conlon, E. Mulrenin, and C. Pohoata, *Two counterexamples to a conjecture about even cycles*, preprint. [[arXiv](#)]
4. E. Mulrenin, *Some remarks on Folkman graphs for triangles*, preprint. [[arXiv](#)]
3. D. Dobák and E. Mulrenin, *Sharp exponents for bipartite Erdős–Rado numbers*, to appear in J. Graph Theory. [[arXiv](#)]
2. E. Mulrenin, C. Pohoata, and D. Zakharov, *Color avoidance for monotone paths*, Discrete Anal. (2025), Paper No. 23, 14 pp. [[arXiv](#)]
1. E. Mulrenin and B. Nagle, *Some imbalanced hypergraph Zarankiewicz numbers*, Bull. Inst. Combin. Appl. **102** (2024), 116–128. [[journal](#)]

TALKS

3. Atlanta lecture series XXX (November 2025); Atlanta, GA, USA
2. University of Georgia analysis seminar (November 2025); Athens, GA, USA
1. Universitat Politècnica de Catalunya combinatorics seminar (November 2024); Barcelona, Spain

¹Updated March 26, 2026

TEACHING EXPERIENCE

Emory University

- Instructor: Calculus I (Fall 2025, Spring 2026)
- Teaching Assistant: Calculus II (Spring 2024), Life Sciences Calculus II (Fall 2024, Spring 2025)
- Grader: Linear Algebra (Fall 2023)

University of South Florida

- Grader: Linear Algebra (Fall 2021, Spring 2022), Introduction to Combinatorics (Fall 2022), Calculus II (Spring 2023)

SERVICE

I have refereed papers for the following journals: *Combinatorica*, *Discrete Mathematics*, *SIAM Journal on Discrete Mathematics*, *Computational Geometry*

Under the Emory mathematics department's directed reading program, I have led the following reading courses:

- Theoretical Computer Science (with undergraduate student Karl Zhou, Spring 2024)
- Philosophy of Mathematics (with undergraduate students Julia Burnham, Catherine Hu, and Edward Liu, Fall 2025)

AWARDS AND SCHOLARSHIPS

- Pi Mu Epsilon Outstanding Scholar Award (2022). *Awarded by the USF Math faculty to one undergraduate student each year for outstanding performance*
- USF College of Arts and Sciences Dean's List (Fall 2019–Spring 2023)
- USF Presidential Scholarship (2019–2023). *\$5000 of financial support per year for four years.*
- Florida Bright Futures Scholarship (2019–2023). *Covers the cost of tuition for eight semesters of undergraduate work.*

ACTIVITIES

- Attended the BIRS workshop on finite geometry and Ramsey theory in Banff, Canada; September 7–12, 2025
- Attended the PCMI graduate summer school in Park City, Utah, USA; July 7–25, 2025
- Attended the SLMATH workshop on algebraic and analytic methods in combinatorics in Berkeley, California, USA; March 16–21, 2025
- Visited Universitat Politècnica de Catalunya in Barcelona, Spain; November 21–27, 2024
- Attended summer school on additive combinatorics at the Rényi Institute in Budapest, Hungary; June 9–14, 2024
- Attended TetFest60 (Prasad Tetali's 60th birthday conference) at Georgia Tech in Atlanta, Georgia, USA; September 9–10, 2023
- Attended the summer school on graph coloring (May 23–26, 2023), the EXCILL IV conference (May 27–28, 2023), and the summer school on algebraic methods in combinatorics (May 29–June 2, 2023) at UIUC in Champaign, Illinois, USA;